

Homework 6: Chi-Square Analysis

Purpose

This assignment helps you move from selecting a hypothesis test to interpreting the evidence it produces. You will review a worked chi-square example, complete two interpretations, and attempt a chi-square analysis connected to your project dataset.

Objectives

By the end of this assignment, you will be able to:

- distinguish a chi-square goodness-of-fit test from a chi-square test of independence;
- interpret a p-value in relation to a stated null hypothesis;
- use standardized residuals and available evidence measures to assess practical importance; and
- complete the five-step hypothesis-testing process for a project-related question.

Resources

- Week 6 and Week 7 slides and interpretation resources
- The four .csv files in the data folder
- The `csf()` and `csi()` functions from `eco230r`
- Your project story and candidate hypotheses

Expectations

This assignment is graded for a sincere analytical attempt, correct use of the five-step process, and interpretations that refer to the variables and evidence in context. Your project-related code may be incomplete, but it must show the test you attempted and any changes you made.

1) Review the Worked Example

The Airbnb dataset is used for the first two hypotheses. Do not change this setup code.

```
df <- read.csv(file.path("data", "airbnb_chicago.csv"))
```

Hypothesis 1 - Worked Example

1. **Statistical story:** What relationship, difference, or pattern are you investigating, and why might it matter?

Airbnb properties are more likely to use a strict cancellation policy than either a flexible or moderate cancellation policy. This may matter to an operator deciding how to balance booking risk with customer flexibility.

2. **Estimate or plot:** What does the descriptive evidence suggest? If there were no uncertainty, what action would the business take?

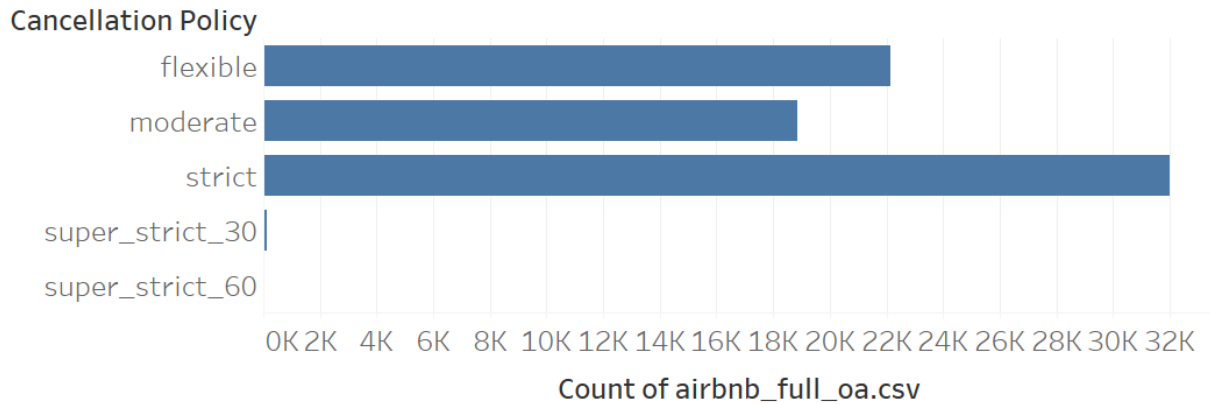


Figure 1: Counts of Airbnb listings by cancellation policy.

The super-strict categories contain very few records, so the analysis below filters them out. Among the remaining categories, strict policies appear most often.

3. **Test and hypotheses:** Which test is appropriate? State H_0 , H_A , and the alpha level.

Test: chi-square goodness-of-fit test using `csf()`

H_0 : Cancellation policy follows the expected model in which flexible, moderate, and strict policies occur equally often.

H_A : Cancellation policy does not follow the expected model.

$\alpha = 0.05$

```
h1 <- df |>
  filter(cancellation_policy %in% c("flexible", "moderate", "strict")) |>
  csf(~ cancellation_policy)

print(h1)
```

4. **Statistical interpretation:** What does the result say about the evidence against H_0 ? Report the relevant estimate and uncertainty in context.

$X^2(2) = 182.008$, $p = 0$

Because the p-value is below 0.05, reject H_0 . The observed cancellation policies do not fit an equal-proportions model.

5. **Practical interpretation:** How large and consequential is the result for a general business audience?

The standardized residuals show which categories contribute most to the difference between the observed and expected counts.

cancellation_policy	Freq
flexible	-9.805714
moderate	-3.121582
strict	12.927297

Strict policies appear more often than expected, while flexible and moderate policies appear less often than expected. An operator could use a more flexible policy to differentiate a listing, but that decision should also account for the

financial risk of cancellations. If the test output provides an effect size or Bayes factor, use it to qualify the magnitude and strength of this conclusion.

2) Complete the Statistical and Practical Interpretations

Hypothesis 2 - Complete Steps 4-5

1. **Statistical story:** What relationship, difference, or pattern are you investigating, and why might it matter?

The number of bathrooms in an Airbnb property is associated with whether the property charges a cleaning fee.

2. **Estimate or plot:** What does the descriptive evidence suggest? If there were no uncertainty, what action would the business take?

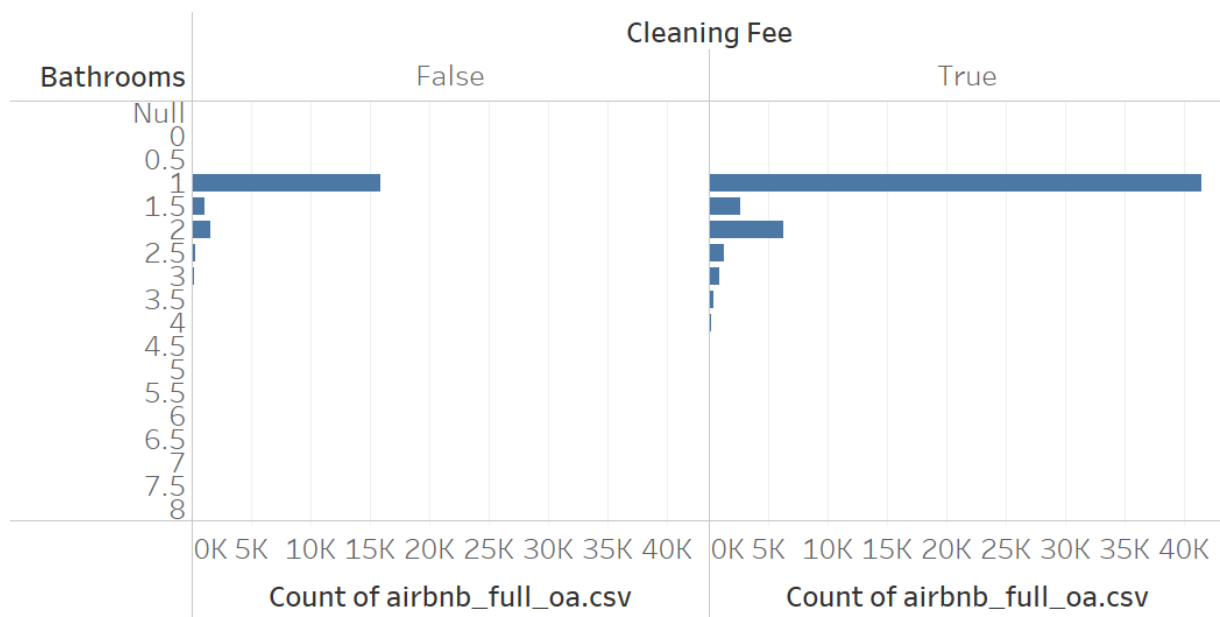


Figure 2: Counts of Airbnb listings by bathroom count and cleaning-fee status.

Most properties have between one and two bathrooms. The test below restricts the analysis to that range so categories with very few records do not drive the result.

3. **Test and hypotheses:** Which test is appropriate? State H_0 , H_A , and the alpha level.

Test: chi-square test of independence using `csi()`

H_0 : Cleaning-fee status is not associated with bathroom category.

H_A : Cleaning-fee status is associated with bathroom category.

$\alpha = 0.05$

```
h2 <- df |>
  filter(bathrooms >= 1, bathrooms <= 2) |>
  mutate(bathroom_category = factor(bathrooms)) |>
  csi(cleaning_fee ~ bathroom_category)

print(h2)
```

4. **Statistical interpretation:** What does the result say about the evidence against H_0 ? Report the relevant estimate and uncertainty in context.

$X^2(2) = 8.844$, $p = 0.012$, $bf_{10} = 0.332$

Your response:

5. **Practical interpretation:** How large and consequential is the result for a general business audience? Use the standardized residuals and table percentages. If your output provides an effect size or Bayes factor, use it to qualify your conclusion.

cleaning_fee	1	1.5	2
FALSE	0.9634757	2.118014	-2.299648
TRUE	-0.9634757	-2.118014	2.299648

Your response:

3) Complete a Project-Related Analysis

Hypothesis 3 - Complete the Full Analysis

Choose the dataset associated with your project. All four options are in the data folder.

Project dataset	Filename
Wisconsin traffic accidents	accident_wi.csv
Chicago Airbnb listings	airbnb_chicago.csv
Wisconsin hospital charges	hospital_charges_wi.csv
Mason Crosby special-teams plays	nfl_special_MasonCrosby.csv

Change project_file below to the filename for your project. Do not change the folder name.

```
project_file <- "accident_wi.csv"
project_data <- read.csv(file.path("data", project_file))

cat(
  project_file, "contains", nrow(project_data), "rows and",
  ncol(project_data), "columns."
)
```

```
accident_wi.csv contains 7919 rows and 47 columns.
```

1. **Statistical story:** What relationship, difference, or pattern are you investigating, and why might it matter?

Your response:

2. **Estimate or plot:** Create or describe at least one estimate or plot. What does the descriptive evidence suggest? If there were no uncertainty, what action would the business or decision maker take?

Your response:

3. **Test and hypotheses:** Select either a chi-square goodness-of-fit test or a chi-square test of independence. State H_0 , H_A , and the alpha level.

Test:

H_0 :

H_A :

$\alpha = 0.05$

Attempt the test below. Replace the placeholder variable names with variables from your project dataset. Keep code that does not yet work, but comment it out before rendering the final PDF.

```
# Example structure for a chi-square goodness-of-fit test:
# h3 <- project_data |>
#   csf(~ categorical_variable)

# Example structure for a chi-square test of independence:
# h3 <- project_data |>
#   csi(outcome_variable ~ group_variable)

# print(h3)
```

4. **Statistical interpretation:** What does the result say about the evidence against H_0 ? If the test did not run, explain the error or decision point you reached.

Your response:

5. **Practical interpretation:** Explain the direction, magnitude, and business consequence of the result. Use effect size or Bayes factor if your output provides them. If the test did not run, explain what evidence you would need to complete this interpretation.

Your response:

Submission

Render this document to PDF and submit Homework_06.pdf in Canvas. Your PDF must contain your responses for Hypotheses 2 and 3 and the code you attempted for Hypothesis 3. Keep Homework_06.qmd in your Posit Cloud project so you can revise it if needed.

Checklist

- ☐ I completed the statistical and practical interpretations for Hypothesis 2.
- ☐ I selected the correct project dataset for Hypothesis 3.
- ☐ I completed all five hypothesis-testing steps for Hypothesis 3.
- ☐ I preserved or commented out the code I attempted.
- ☐ I rendered the document successfully and reviewed the PDF.
- ☐ I submitted Homework_06.pdf in Canvas.

Grading focus: Your work will be evaluated for correct hypothesis-test structure, use of evidence in context, distinction between statistical and practical importance, and a sincere project-related analysis attempt.